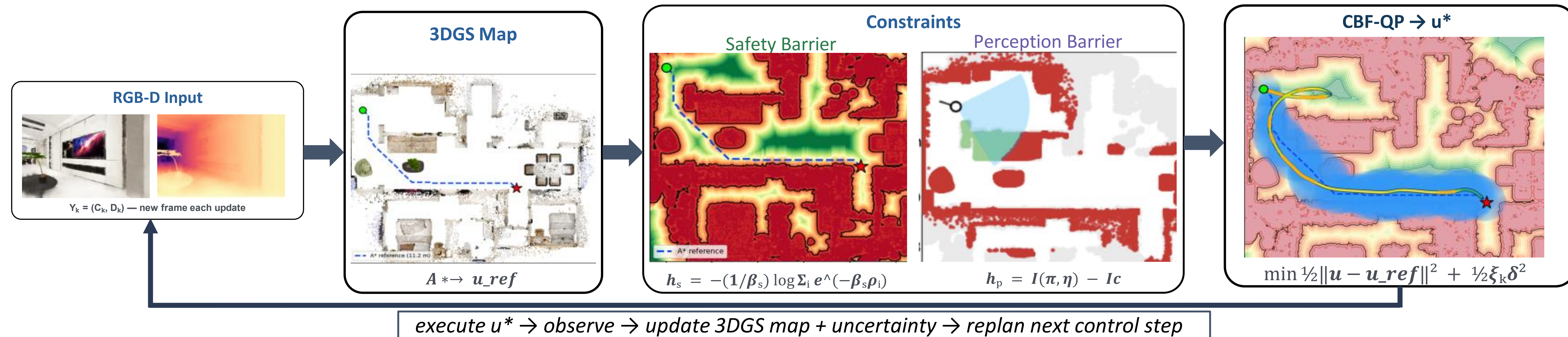


System Overview

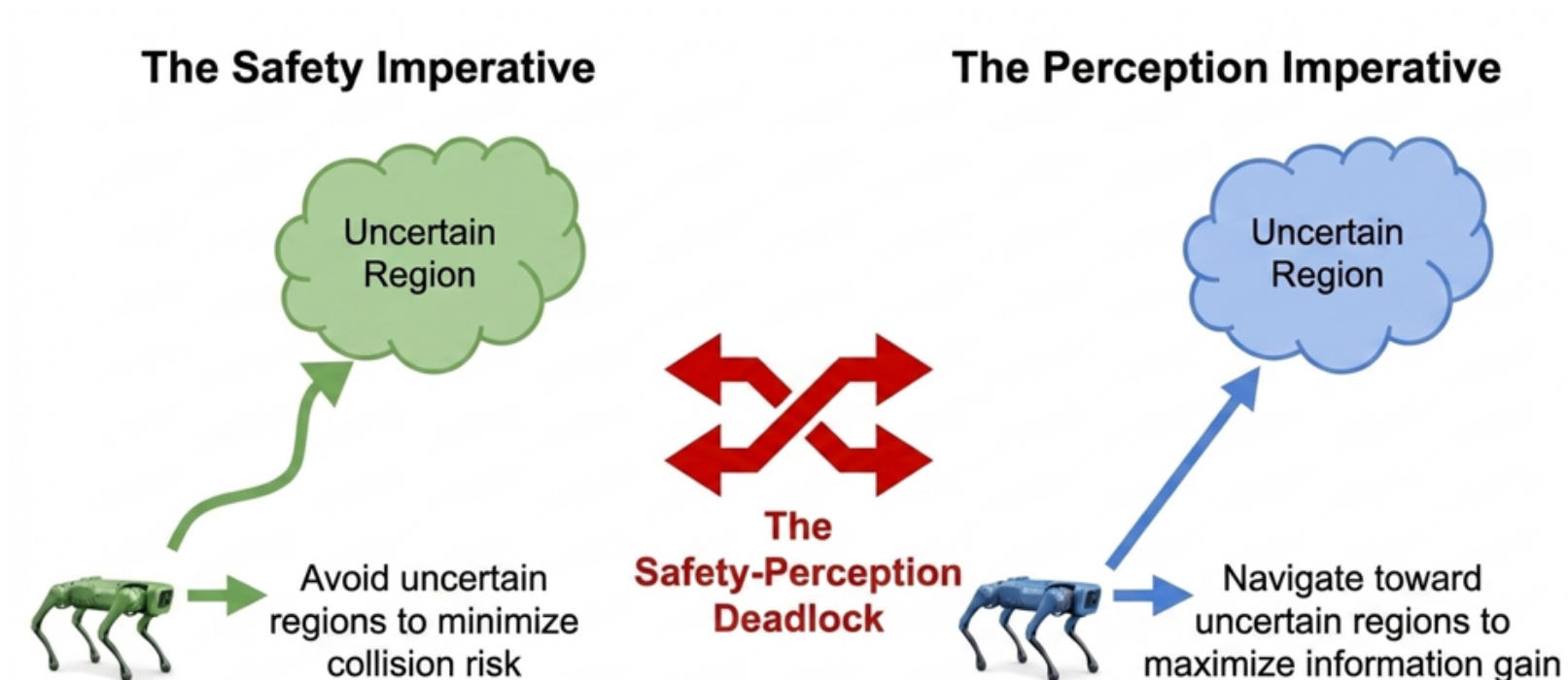


- 1. Observe.** An RGB-D view updates the **online 3DGS map**.
- 2. Evaluate.** A semantic risk-aware barrier certifies clearance from nearby Gaussians; a perception barrier scores information along the near-future trajectory.
- 3. Resolve.** One QP modifies the nominal command: **safety carries no slack, perception may yield**.
- 4. Act and update.** Execute the safe next-best-view action, acquire a new observation, repeat.

■ objectives compatible: $\delta_p = 0$ ■ perception relaxed: $\delta_p > 0$

1. Problem Formulation

A robot must reach a goal while its 3DGS map is uncertain and the views that resolve that uncertainty look straight into the geometry safety most wants to avoid.



To be safe, a robot must **avoid** uncertainty.
To resolve uncertainty, it must **look straight at it**.

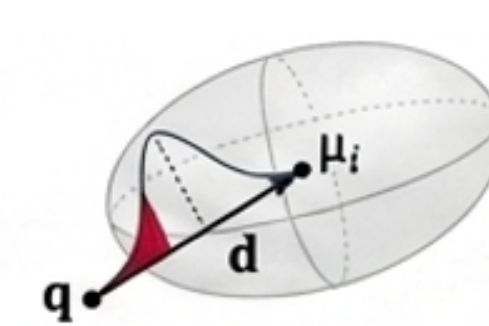
Our solution contributes:

- one smooth, semantically weighted safety certificate;
- trajectory-aware Fisher information as a perception barrier;
- an adaptive-slack QP that gives safety strict priority.

2. Safety Barrier

Enforced with no slack variable — it can never be traded away.

Model the robot-to-splat distance with each Gaussian's geometric uncertainty and protect against its **adverse tail**.



A positive **semantic risk weight** $\kappa(\xi_i)$ maps each splat's semantic label and confidence to relative collision risk. Safety-critical or semantically uncertain objects receive *smaller* κ , shrinking their effective clearance and raising their pull on the barrier.

$$d_i(\pi) \sim \mathcal{N}(\|\mu_i - \pi\|_2, \sigma_i^2), \quad \rho_i^{\text{sem}}(\pi) = \kappa(\xi_i)(\text{AV@R}_\varepsilon[d_i(\pi)] - r_{\text{rob}}).$$

Map-wide safety needs $\min_i \rho_i^{\text{sem}} \geq 0$, which is nonsmooth. Collapse the active set $\mathcal{A}_k$ into *one* differentiable barrier:

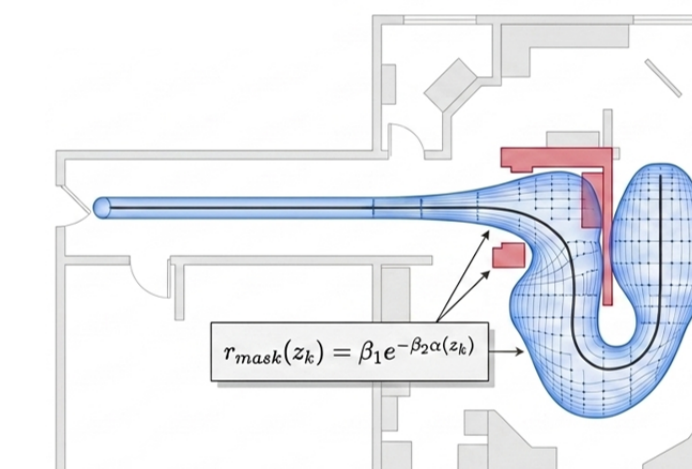
$$h_s(\pi) = -\frac{1}{\beta_s} \log \sum_{g \in \mathcal{A}_k} e^{-\beta_s \rho_i^{\text{sem}}(\pi)}$$

A log-sum-exp sandwich bound makes $h_s \geq 0$ **sufficient** for nonnegative semantic risk-adjusted clearance from every active Gaussian — one constraint, no switching, no per-splat rows.

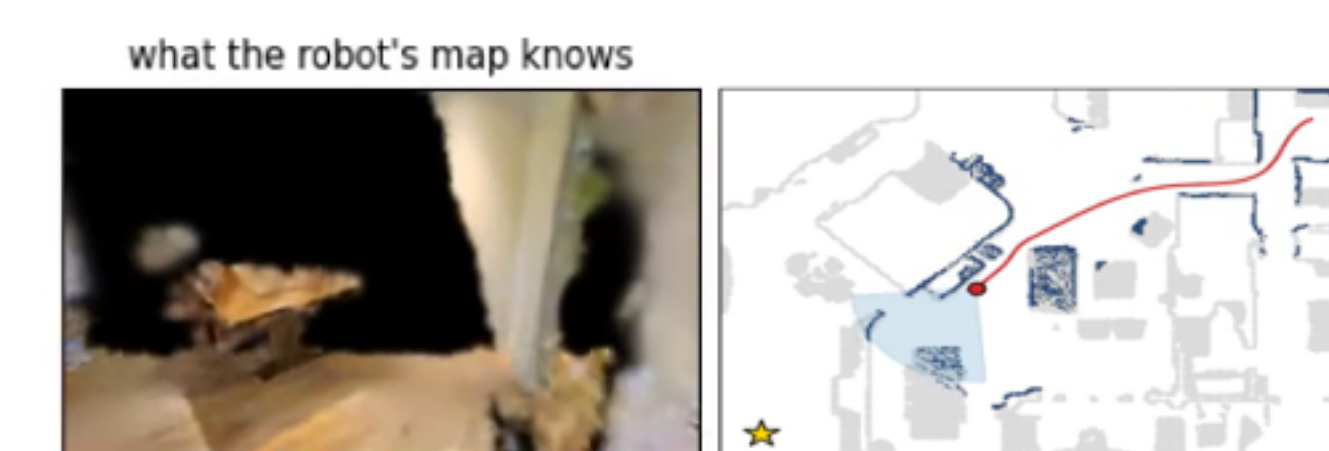
3. Perception Barrier

Carries a slack $\delta_p \geq 0$ — it yields, at a price, only when safety demands it.

Do not improve the map everywhere — improve the part the robot is about to drive through. A risk-inflated tube around the nominal rollout selects the path-relevant splats $\mathcal{G}_{P_k}$, widening where clearance is thin.



Path-relevant tube $\mathcal{G}_{P_k}$



Onboard view being scored

A view (π, η) is scored by how much it would shrink the remaining uncertainty of those splats (diagonal Fisher / Gauss-Newton):

$$\tilde{H}(\pi, \eta) = \text{diag}(J_w^T J_w) + \lambda I, \quad I(\pi, \eta) \approx \text{tr}(\tilde{H} H_{\text{prior}}^{-1}).$$

Demanding a *minimum* score turns information from a hand-tuned reward into a constraint the controller must justify breaking:

$$h_p(\pi, \eta) = I(\pi, \eta) - I_c$$

Sensing lands on the uncertainty that can actually limit future motion.

The Unified CBF-QP: Safety Never Yields; Perception Yields Only When Necessary

$$\begin{aligned} \min_{u_k, \delta_p \geq 0} \quad & \frac{1}{2} \|u_k - u_k^{\text{ref}}\|_2^2 + \frac{1}{2} w_p(x_k) \delta_p^2 \\ \text{s.t.} \quad & \dot{h}_s \geq -\alpha_s(h_s) \\ & \dot{h}_p \geq -\alpha_p(h_p) - \delta_p \\ & u_{\min} \leq u_k \leq u_{\max}. \end{aligned}$$

no slack
slack δ_p

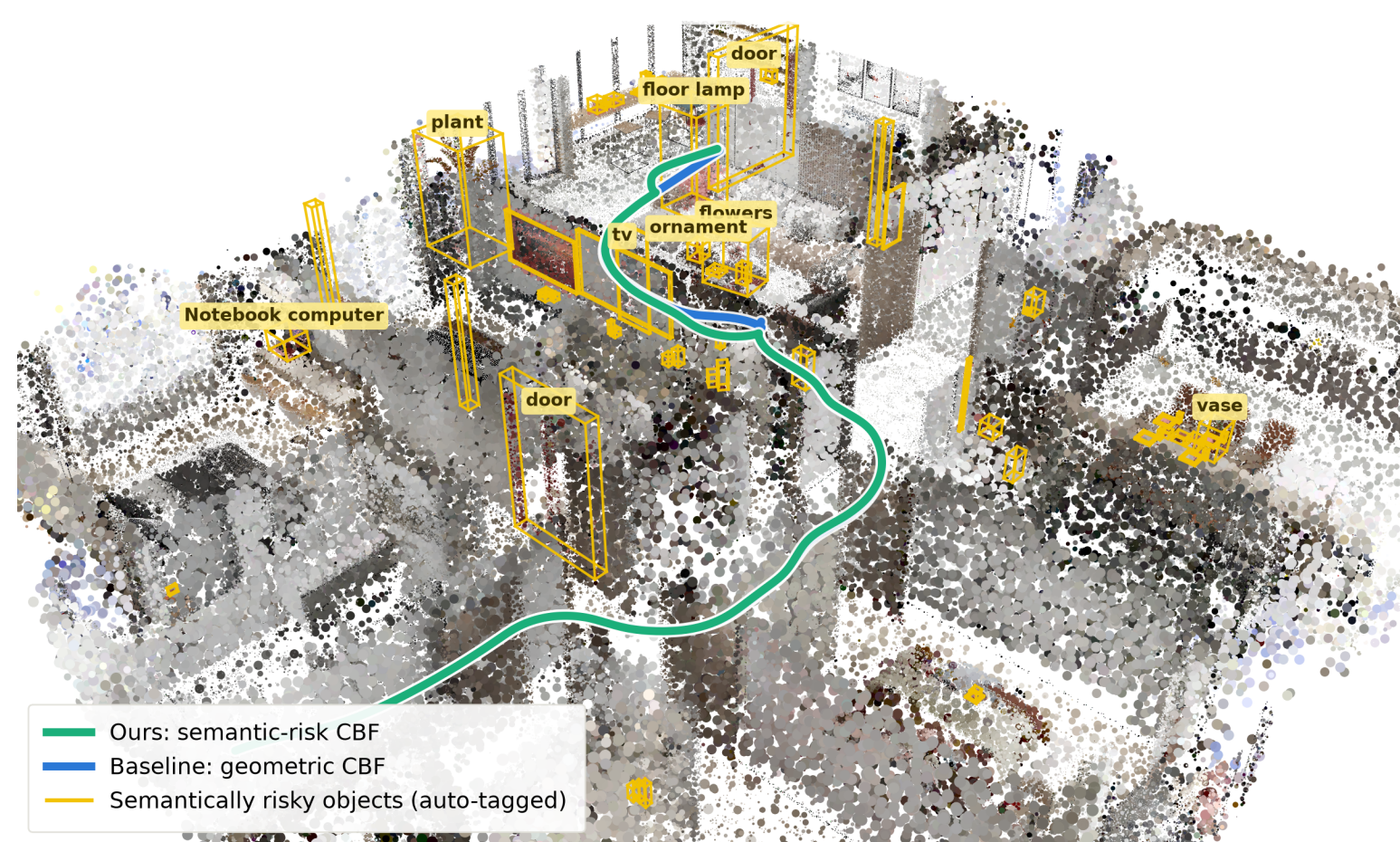
The safety row carries **no slack variable**: its multiplier is uncapped, so it can never be traded. KKT gives $\lambda_p = w_p \delta_p^*$ — the weight is the **marginal price of lost information**, adapted online:

Persistent slack + sufficient margin $\Rightarrow w_p \uparrow$

Near the safety boundary $\Rightarrow w_p \downarrow$

It keeps looking while that is safe, and restores safety priority at once near obstacles.

Safety & Speed



Semantic risk-aware navigation in a 3DGS home.

One barrier instead of one per splat. Four 3DGS scenes, each 100 trials:

Method	Safety [%] ↑	Success [%] ↑	Time [ms] ↓	Min. dist. [m] ↑
PolyMerge [†]	96.8	95.5	1.16	1.53
SAFER-Splat	99.5	88.5	63.82	0.86
SemSafe $\varepsilon=.30$	99.5	93.3	0.99	1.06
SemSafe $\varepsilon=.10$	99.5	83.5	1.00	1.43

Semantics buy clearance where it matters. Same route, same map:

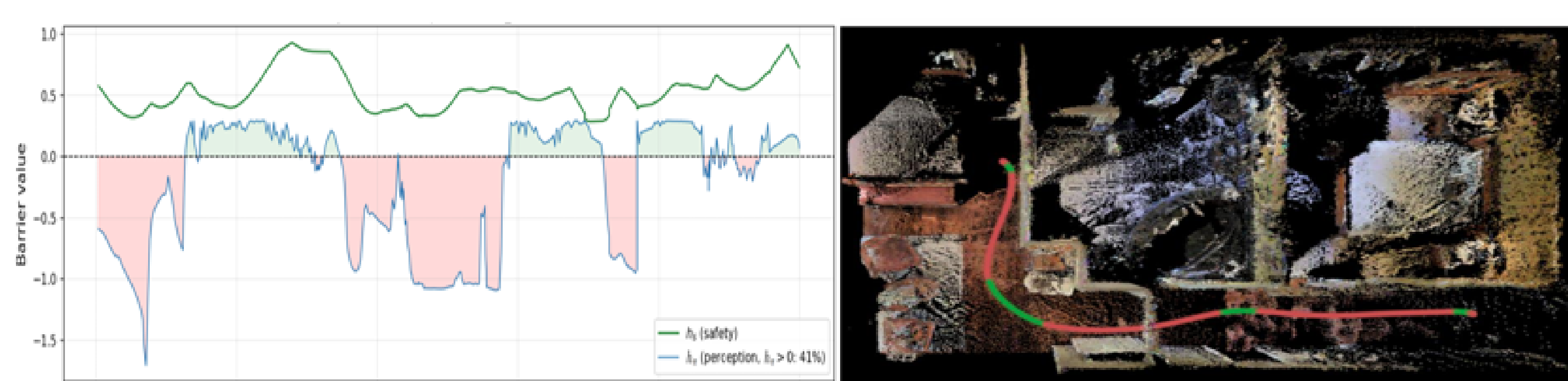
Controller	Time [s]	min h_s [m]	High-risk clr. [cm]
Geometric	59.8	0.026	14.8
Semantic	86.2	0.033	36.7

2.5× the clearance from high-risk objects at identical path length (20.95 vs. 20.97 m).

64× faster online control than SAFER-Splat, at matched safety.

[†]PolyMerge offline compression excluded from online timing.

Active Perception



Same start and goal in InteriorGS. Passive mapping stays conservative under unresolved uncertainty.

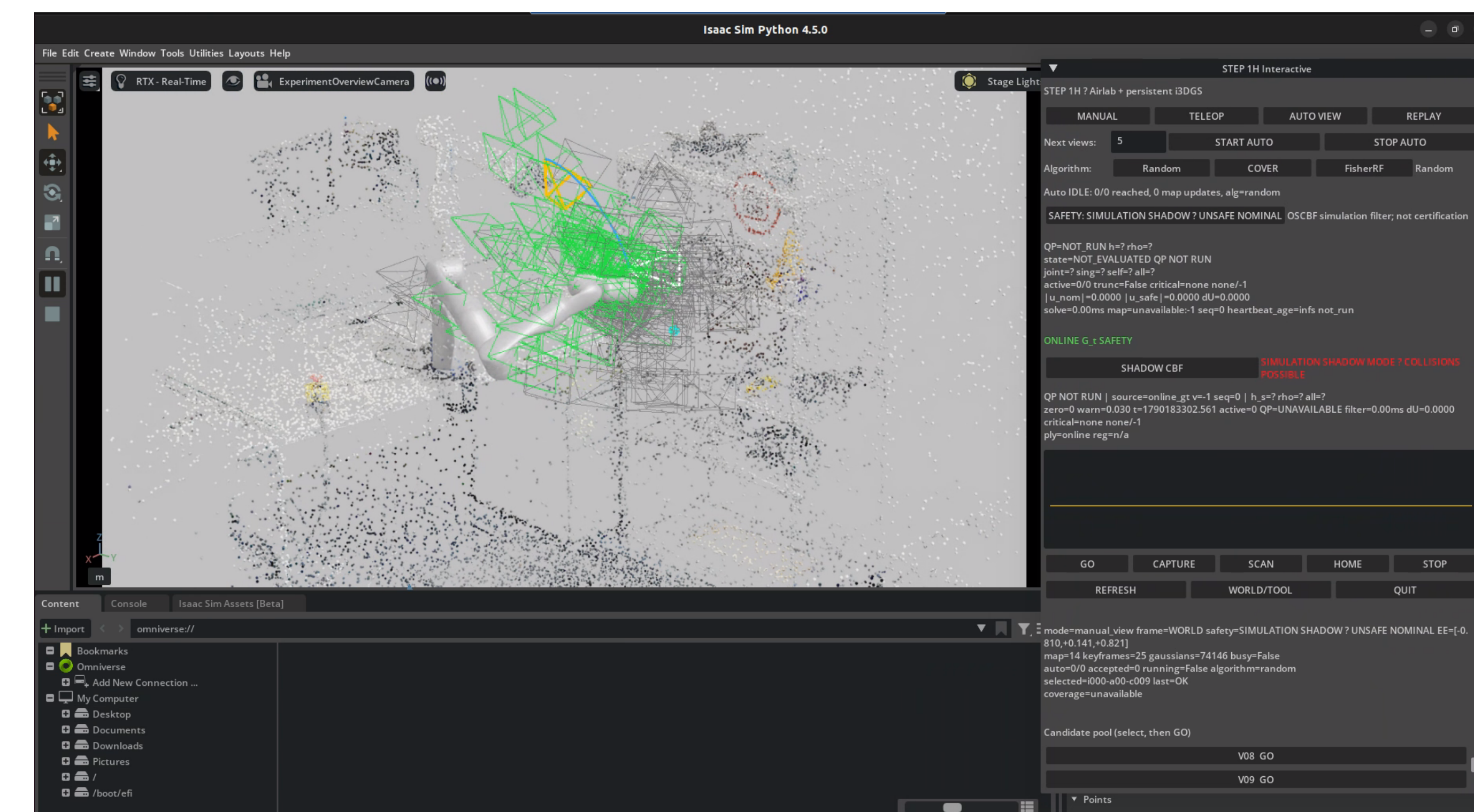
	Ours	CAAP	Passive
Goal time [s] ↓	74	82	103
Path length [m]	16.6	16.9	16.4
Mean speed [m/s] ↑	0.212	0.192	0.143
Perception active ↑	41%	33%	—

How a conflict is resolved

- **Compatible.** Both barriers hold, $\delta_p = 0$; the QP returns the informative action closest to u^{ref} .
- **In conflict.** An informative view would eat the clearance margin. Safety is held *exactly*; δ_p absorbs the perception violation at cost $w_p \delta_p^2$.
- **Recovered.** Margin returns, $\delta_p \rightarrow 0$, informative motion resumes.

Safety is never traded — perception is only deferred.

Manipulator & Real Robot



Kinova Gen3 in Immediate3DGS and Ackermann hardware validation.

	COVER	PSNR ↑	SSIM ↑	$h_{\text{env}}^{\min} \uparrow$
Safety only	17.08	0.712	0.089	
+ SemSafe	17.96	0.737	0.138	

Better reconstruction and a larger environment margin — for every view selector.

Kinova Gen3. All trials collision-free; median QP solve 1.13–1.18 ms with the perception constraint active.

Ackermann. $h_s > 0$ throughout; conflicts occupy 24.2% of the run with adaptive $w_p \in [0.02, 0.49]$; 91% final map coverage.

One hierarchy across unicycle, Ackermann, and 7-DoF arm dynamics.